

Homework 5: System F

The appendix gives the formalization of untyped lambda calculus and System F.

The problems below consider type erasure, a translation $|M|$ from System F terms to untyped terms that removes type annotations, type abstractions, and type applications:

$$\begin{aligned} |x| &= x, & |\lambda x : \tau. M| &= \lambda x. |M|, & |M N| &= |M| |N|, \\ |\Lambda \alpha. M| &= |M|, & |M \langle \sigma \rangle| &= |M|. \end{aligned}$$

1. Typing, translation, and reduction.

For each System F term below:

- (i) Find its type and give a complete typing derivation in the empty contexts.
- (ii) Give its translation into the untyped lambda calculus, without reduction.
- (iii) Write the full reduction sequences to normal form for both the original term and its translation. Use leftmost outermost reduction.

You may name and reuse subderivations and cite type well-formedness judgments without deriving them.

(a)

$$\begin{aligned} &(\Lambda \alpha. \Lambda \beta. \Lambda \gamma. \lambda f : \alpha \rightarrow \beta \rightarrow \gamma. \lambda y : \beta. \lambda x : \alpha. f \ x \ y) \\ &\langle T \rightarrow T \rangle \langle T \rangle \langle T \rangle (\lambda g : T \rightarrow T. \lambda z : T. g \ z) \end{aligned}$$

(b)

$$\begin{aligned} &(\Lambda \gamma. \lambda k : (T \rightarrow T) \rightarrow (T \rightarrow T \rightarrow T) \rightarrow \gamma. \\ &\quad k \ (\lambda z : T. z) (\lambda u : T. \lambda v : T. u)) \\ &\langle T \rightarrow T \rightarrow T \rangle (\lambda f : T \rightarrow T. \lambda g : T \rightarrow T \rightarrow T. g) \end{aligned}$$

2. Properties of type erasure.

Does each of the following properties hold? If it holds, just answer yes; no proof is required. Otherwise, give a counterexample and explain why it violates the property.

- (a) If $|M|$ is an untyped normal form, then M is a System F normal form.
- (b) If M is a System F normal form, then $|M|$ is an untyped normal form.
- (c) If M is well typed and $|M| = |N|$, then N is also well typed. Here we say a term M is well typed if there exists some type τ such that $\bullet; \bullet \vdash M : \tau$.

Appendix

A Untyped Lambda Calculus

Syntax:

$$\begin{aligned} \text{(Terms)} \quad U, V &::= x \mid \lambda x. U \mid U V \\ \text{(Values)} \quad v &::= \lambda x. U \end{aligned}$$

Free variables:

$$\begin{aligned} fv(x) &= \{x\}, \\ fv(U V) &= fv(U) \cup fv(V), \\ fv(\lambda x. U) &= fv(U) \setminus \{x\}. \end{aligned}$$

Substitution (here $x \neq y$):

$$\begin{aligned} x[V/x] &= V, \\ y[V/x] &= y, \\ (U_1 U_2)[V/x] &= (U_1[V/x]) (U_2[V/x]), \\ (\lambda x. U)[V/x] &= \lambda x. U, \\ (\lambda y. U)[V/x] &= \lambda y. (U[V/x]) && \text{if } y \notin fv(V), \\ (\lambda y. U)[V/x] &= \lambda z. ((U[z/y])[V/x]) && \text{if } y \in fv(V), \text{ } z \text{ fresh.} \end{aligned}$$

Reduction rules:

$$\begin{aligned} \frac{}{(\lambda x. U) V \longrightarrow_u U[V/x]} \quad & \frac{U \longrightarrow_u U'}{\lambda x. U \longrightarrow_u \lambda x. U'} \\ \frac{U \longrightarrow_u U'}{U V \longrightarrow_u U' V} \quad & \frac{V \longrightarrow_u V'}{U V \longrightarrow_u U V'} \end{aligned}$$

Multistep reduction ($k \geq 0$):

$$\begin{aligned} U \longrightarrow_u^0 V &\quad \text{iff } U = V, \\ U \longrightarrow_u^{k+1} V &\quad \text{iff } \exists W. U \longrightarrow_u W \wedge W \longrightarrow_u^k V, \\ U \longrightarrow_u^* V &\quad \text{iff } \exists k \geq 0. U \longrightarrow_u^k V. \end{aligned}$$

Normal form: a term U for which no U' satisfies $U \longrightarrow_u U'$.

Closed term: a term U with $fv(U) = \emptyset$.

B System F

Syntax:

$$\begin{aligned} \text{(Terms)} \quad M, N &::= x \mid \lambda x : \tau. M \mid M N \mid \Lambda \alpha. M \mid M \langle \tau \rangle \\ \text{(Types)} \quad \tau, \sigma &::= \alpha \mid \mathbf{T} \mid \tau \rightarrow \tau \mid \forall \alpha. \tau \\ \text{(Values)} \quad v &::= \lambda x : \tau. M \mid \Lambda \alpha. M \\ \text{(Contexts)} \quad \Gamma &::= \bullet \mid \Gamma, x : \tau \\ \text{(TypeVarContexts)} \quad \Delta &::= \bullet \mid \Delta, \alpha \end{aligned}$$

Context extensions use fresh names; all types in Γ are well formed under Δ .

Type well-formedness: $\Delta \vdash \tau$.

$$\frac{\alpha \in \Delta}{\Delta \vdash \alpha} \quad \frac{}{\Delta \vdash \mathbf{T}}$$

$$\frac{\Delta \vdash \sigma \quad \Delta \vdash \tau}{\Delta \vdash \sigma \rightarrow \tau} \quad \frac{\Delta, \alpha \vdash \tau}{\Delta \vdash \forall \alpha. \tau}$$

Typing rules: $\Delta; \Gamma \vdash M : \tau$.

$$\begin{aligned} & \frac{x : \tau \in \Gamma}{\Delta; \Gamma \vdash x : \tau} \text{ (T-VAR)} \\ & \frac{\Delta \vdash \sigma \quad \Delta; \Gamma, x : \sigma \vdash M : \tau}{\Delta; \Gamma \vdash \lambda x : \sigma. M : \sigma \rightarrow \tau} \text{ (T-ABS)} \\ & \frac{\Delta; \Gamma \vdash M : \sigma \rightarrow \tau \quad \Delta; \Gamma \vdash N : \sigma}{\Delta; \Gamma \vdash M N : \tau} \text{ (T-APP)} \\ & \frac{\Delta, \alpha; \Gamma \vdash M : \tau}{\Delta; \Gamma \vdash \Lambda \alpha. M : \forall \alpha. \tau} \text{ (T-TABS)} \\ & \frac{\Delta; \Gamma \vdash M : \forall \alpha. \tau \quad \Delta \vdash \sigma}{\Delta; \Gamma \vdash M \langle \sigma \rangle : \tau[\sigma/\alpha]} \text{ (T-TAPP)} \end{aligned}$$

Reduction rules:

$$\begin{aligned} & \overline{(\lambda x : \tau. M) N \longrightarrow_F M[N/x]} \text{ (E-APPABS)} \\ & \frac{M \longrightarrow_F M'}{M N \longrightarrow_F M' N} \text{ (E-APP1)} \quad \frac{N \longrightarrow_F N'}{M N \longrightarrow_F M N'} \text{ (E-APP2)} \\ & \frac{M \longrightarrow_F M'}{\lambda x : \tau. M \longrightarrow_F \lambda x : \tau. M'} \text{ (E-ABS)} \\ & \overline{(\Lambda \alpha. M) \langle \sigma \rangle \longrightarrow_F M[\sigma/\alpha]} \text{ (E-TAPPTABS)} \\ & \frac{M \longrightarrow_F M'}{M \langle \tau \rangle \longrightarrow_F M' \langle \tau \rangle} \text{ (E-TAPP)} \quad \frac{M \longrightarrow_F M'}{\Lambda \alpha. M \longrightarrow_F \Lambda \alpha. M'} \text{ (E-TABS)} \end{aligned}$$

Normal form: a term M for which no M' satisfies $M \longrightarrow_F M'$.